

# Daily Sequences Drill #1 — Answers & Investigations

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| Section | Topic                    | Questions | Target time  |
|---------|--------------------------|-----------|--------------|
| A       | Rapid Recognition        | 10        | 2 min 30 sec |
| B       | Manipulation Drills      | 10        | 8 min        |
| C       | Substitution & Structure | 8         | 10 min       |
| D       | Challenge                | 5         | 15 min       |

New toolkit today: Arithmetic Sequences & Series ·  $n$ th-term & Sum Formulas ·  $S_n$  as a Quadratic in  $n$ .

## Section A Rapid Recognition

1. Write down the  $n$ th term of the arithmetic sequence 5, 8, 11, 14, ...

**Answer:**  $3n + 2$

**Method:**  $a = 5$ ,  $d = 3$ .  $a_n = a + (n - 1)d = 5 + 3(n - 1) = 3n + 2$ .

**Investigate further:** Check:  $a_1 = 5$  ✓,  $a_4 = 14$  ✓. Every AP's  $n$ th term is linear in  $n$  — the coefficient of  $n$  is always the common difference.

2. An AP has first term 7 and common difference  $-3$ . Find the 20th term.

**Answer:**  $-50$

**Method:**  $a_{20} = 7 + 19(-3) = 7 - 57 = -50$ .

**Investigate further:** A negative common difference just means the “ $n - 1$  steps” subtract instead of add. The formula never changes.

3. An AP has first term 4 and common difference 5. Find the sum of the first 10 terms.

**Answer:** 265

**Method:**  $S_{10} = \frac{10}{2}(2(4) + 9(5)) = 5(8 + 45) = 5(53) = 265$ .

**Investigate further:** Cross-check with  $S_n = \frac{n}{2}(a_1 + a_n)$ :  $a_{10} = 4 + 9(5) = 49$ ,  $S_{10} = 5(4 + 49) = 265$ . ✓

4. A sequence has  $n$ th term  $4n - 1$ . Write down its first three terms, and state whether the sequence is arithmetic.

**Answer:** 3, 7, 11; arithmetic with  $d = 4$

**Method:**  $a_1 = 3$ ,  $a_2 = 7$ ,  $a_3 = 11$ . Consecutive differences are both 4, so the sequence is arithmetic.

**Investigate further:** Any  $n$ th term of the form  $pn + q$  is automatically arithmetic with  $d = p$  — this is worth recognising instantly rather than re-deriving each time.

5. An AP has  $a_1 = 2$  and  $a_5 = 18$ . Find the common difference.

**Answer:** 4

**Method:**  $a_5 - a_1 = 4d = 16 \Rightarrow d = 4$ .

**Investigate further:** In general,  $a_m - a_k = (m - k)d$  — the number of “steps” between any two terms is the index difference, not one of the indices themselves.

6. Evaluate  $1 + 2 + 3 + \cdots + 50$ .

**Answer:** 1275

**Method:**  $S_{50} = \frac{50 \cdot 51}{2} = 1275$ .

**Investigate further:** This is Gauss’s schoolroom trick: pair the 1st and 50th, 2nd and 49th, etc. — each pair sums to 51, and there are 25 pairs.

7. An AP has  $S_n = n^2 + 3n$ . Find the first term  $a_1$ .

**Answer:** 4

**Method:**  $a_1 = S_1 = 1^2 + 3(1) = 4$ .

**Investigate further:** More generally  $a_n = S_n - S_{n-1}$  for  $n \geq 2$ , but  $a_1$  is just  $S_1$  on its own — a common slip is applying the subtraction formula to  $n = 1$  too, which needs  $S_0$  (always 0) instead.

8. True or false: every arithmetic sequence with integer first term and integer common difference consists entirely of integers.

**Answer:** True

**Method:**  $a_n = a_1 + (n - 1)d$  is a sum of integers whenever  $a_1, d, n$  are integers, so every term is an integer.

**Investigate further:** The converse is false: a sequence of all integers need not be arithmetic (e.g.  $1, 2, 4, 8, \dots$ ).

9. An AP has 3rd term 13 and 9th term 37. Find the common difference.

**Answer:** 4

**Method:** 9th term – 3rd term is 6 steps of  $d$ :  $37 - 13 = 24 = 6d \Rightarrow d = 4$ .

**Investigate further:** Always count steps as (index difference), not (larger index). Between the 3rd and 9th term there are  $9 - 3 = 6$  gaps, not 9 or 3.

10. An AP has first term 4 and 6th term 29. Find the common difference.

**Answer:** 5

**Method:** 6th term – 1st term is 5 steps:  $29 - 4 = 25 = 5d \Rightarrow d = 5$ .

**Investigate further:** Same step-counting idea as A9, now anchored at the first term directly.

## Section B Manipulation Drills

1. An AP has first term 3, last term 59, and 15 terms. Find the common difference and the sum of all 15 terms.

**Answer:**  $d = 4$ ,  $S_{15} = 465$

**Method:**  $l = a + (n - 1)d \Rightarrow 59 = 3 + 14d \Rightarrow d = 4$ .  $S_{15} = \frac{15}{2}(3 + 59) = \frac{15}{2}(62) = 465$ .

**Investigate further:** Whenever you're given first term, last term, and count,  $S_n = \frac{n}{2}(a + l)$  is faster than finding  $d$  first and building back up through  $S_n = \frac{n}{2}(2a + (n - 1)d)$ .

2. The sum of the first 8 terms of an AP is 100, and the sum of the first 4 terms is 30. What is the sum of terms 5 through 8?

- A) 50  
B) 70  
C) 130  
D) 100

**Answer:** B) 70

**Method:**  $S_8 - S_4$  is exactly the sum of terms 5 through 8, by definition of  $S_n$  as a running total:  $100 - 30 = 70$ . No need to ever find  $a$  or  $d$ .

**Investigate further:** The trap is trying to solve for  $a, d$  from the two sums first — possible, but a detour.  $S_8 - S_4$  is a direct subtraction of two given numbers.

3. Prove that  $3 + 6 + 9 + \cdots + 3n = \frac{3n(n+1)}{2}$  using the arithmetic series sum formula.

**Answer:** Proof by the AP sum formula.

**Method:**  $3, 6, 9, \dots, 3n$  is an AP with  $a = 3$ ,  $d = 3$ , last term  $3n$ , and  $n$  terms.  $S_n = \frac{n}{2}(a + l) = \frac{n}{2}(3 + 3n) = \frac{3n(n+1)}{2}$ .  $\square$

**Investigate further:** This is exactly  $3 \times (1 + 2 + \cdots + n)$  — scaling every term of an AP by a constant  $k$  scales the sum by  $k$  too, since  $S_n$  is linear in the terms.

4. An AP has 21 terms summing to 315. Find the middle (11th) term.

- A) 15  
B) 315  
C) 21  
D) It cannot be determined without knowing  $a$  and  $d$ .

**Answer:** A) 15

**Method:** For an AP with an odd number of terms  $n = 2k + 1$ , the sum equals  $n$  times the middle term (the terms pair up symmetrically around it, and the middle term is left over unpaired but equal to the pair-average). Here  $315 = 21 \times (\text{middle term}) \Rightarrow \text{middle term} = 15$ .

**Investigate further:** This only works because 21 is odd — for an even-length AP there's no single middle term, and you'd need  $S_n = \frac{n}{2}(a_1 + a_n)$  instead, which needs both endpoints.

5. An AP has first term 10 and common difference  $-3$ . Find the last term of the sequence that is still positive.

**Answer:** 1 (the 4th term)

**Method:**  $a_n = 10 - 3(n - 1) > 0 \Rightarrow n < \frac{13}{3} \approx 4.33$ , so the last positive term is  $a_4 = 10 - 9 = 1$  (and  $a_5 = -2 < 0$  confirms the sequence has already crossed zero).

**Investigate further:** Always check the boundary term itself, not just the inequality's cutoff —  $n < 4.33$  tells you  $n = 4$  is the last valid integer, but you still need to confirm  $a_4$  is actually positive (it could in principle have landed exactly on 0).

6. Which of the following is NOT necessarily true for an arithmetic sequence with common difference  $d > 0$ ?

- A) The sequence is strictly increasing.
- B)  $a_{n+1} - a_n = d$  for every  $n$ .
- C) The sequence is unbounded above.
- D) All terms are positive.

**Answer:** D) All terms are positive.

**Method:**  $d > 0$  guarantees the sequence is strictly increasing (A, true) and therefore unbounded above (C, true), and by definition  $a_{n+1} - a_n = d$  always (B, true). But the first term  $a_1$  can be any real number, including very negative — e.g.  $a_1 = -100, d = 1$  has plenty of negative terms. So D is the one that need not hold.

**Investigate further:**  $d > 0$  constrains the *shape* of the sequence, not its starting position.

7. Two APs share the same common difference  $d$ . Their first terms differ by 6. What is the difference between their  $n$ th terms, for any  $n$ ?

- A) 6
- B)  $6 + (n - 1)d$
- C)  $6n$
- D) It depends on  $d$ .

**Answer:** A) 6

**Method:** Let the sequences be  $a_n = a_1 + (n - 1)d$  and  $b_n = b_1 + (n - 1)d$  (same  $d$ ). Then  $a_n - b_n = a_1 - b_1 = 6$  for every  $n$  — the  $(n - 1)d$  terms cancel exactly because the common difference is shared.

**Investigate further:** If the common differences were different, the gap would grow (or shrink) linearly in  $n$  instead of staying constant — try it with  $d_1 = 2, d_2 = 3$  to see the gap become  $6 + (n - 1)$ .

8. An AP has  $a_3 = 11$  and  $a_3 + a_5 = 32$ . Find the common difference.

- A) 3
- B) 4
- C) 5
- D) 6

**Answer:** C) 5

**Method:**  $a_3 + a_5 = (a + 2d) + (a + 4d) = 2a + 6d = 32 \Rightarrow a + 3d = 16$ . Subtracting  $a_3 = a + 2d = 11$ :  $(a + 3d) - (a + 2d) = 16 - 11 \Rightarrow d = 5$ .

**Investigate further:** Once  $d = 5$ , back-substitute:  $a = 11 - 2(5) = 1$ . Always sanity-check by plugging both original conditions back in:  $a_3 = 1 + 10 = 11 \checkmark$ ,  $a_3 + a_5 = 11 + 21 = 32 \checkmark$ .

9. An AP has first term  $a$  and common difference  $d$ , both positive integers, and its 5th term equals 17. Which pair  $(a, d)$  is NOT possible?

- A)  $(13, 1)$
- B)  $(9, 2)$
- C)  $(5, 3)$
- D)  $(2, 4)$

**Answer:** D)  $(2, 4)$

**Method:** Need  $a + 4d = 17$  with  $a, d$  positive integers. Check each:  $13 + 4(1) = 17 \checkmark$ ,  $9 + 4(2) = 17 \checkmark$ ,  $5 + 4(3) = 17 \checkmark$ , but  $2 + 4(4) = 18 \neq 17$  — D fails the defining equation outright.

**Investigate further:** The three valid pairs follow the pattern “ $a$  drops by 4 as  $d$  rises by 1”  $(13, 9, 5, 1, \dots)$  — option D’s  $a = 2$  breaks that pattern, which is worth noticing as a sanity check even before computing.

10. An AP with first term 2 and common difference 3 (terms  $2, 5, 8, 11, \dots$ ) and a GP with first term 2 and common ratio 2 (terms  $2, 4, 8, 16, \dots$ ) are compared term by term. At which term number do they next agree, within the first 6 terms of each?

- A)  $n = 2$
- B)  $n = 3$
- C)  $n = 4$
- D) They never agree again within 6 terms.

**Answer:** B)  $n = 3$

**Method:** AP:  $2, 5, 8, 11, 14, 17$ . GP:  $2, 4, 8, 16, 32, 64$ . Comparing term by term, both give 8 at the 3rd term.

**Investigate further:** After  $n = 3$  the GP overtakes the AP permanently and they never meet again — geometric growth eventually dominates linear growth no matter the starting ratio, a theme this pillar returns to when comparing arithmetic and geometric series directly.

## Section C Substitution & Structure

1. An AP has  $n$  integer terms ( $n \geq 2$ ) summing to 15. Which of the following must be true? (after TMUA 2019 Paper 2 Q11)

- I.  $n$  is even.
  - II. The first term is odd.
  - III. The common difference is odd.
- A) none of them
  - B) I only
  - C) II only

- D) III only
- E) I and II only
- F) I and III only
- G) II and III only
- H) I, II and III

**Answer:** A) none of them

**Method:** Writing  $S_n = \frac{n}{2}(2a + (n-1)d) = 15$  gives  $n(2a + (n-1)d) = 30$ , so  $n$  must divide 30. Counterexamples kill every statement at once: with  $n = 3, a = 4, d = 1$  the sequence is 4, 5, 6 (sum 15) —  $n = 3$  is odd (kills I) and  $a = 4$  is even (kills II). With  $n = 3, a = 3, d = 2$  the sequence is 3, 5, 7 (sum 15) —  $d = 2$  is even (kills III). No single parity pattern is forced.

**Investigate further:** This is the real trap of “must be true” questions: checking one valid example never proves a statement false, but finding *one* counterexample per statement is enough to rule each one out individually.

2. The sequences  $a_n = 2n + 1$  and  $b_n = 3n - 2$  are both arithmetic. Let  $c_n = a_n + b_n$ . Which is true about  $(c_n)$ ?

- A) It is arithmetic with common difference 5.
- B) It is arithmetic with common difference 6.
- C) It is not arithmetic in general.
- D) It is geometric.

**Answer:** A) arithmetic with common difference 5

**Method:**  $c_n = (2n + 1) + (3n - 2) = 5n - 1$ , which is linear in  $n$  with slope 5 — so  $(c_n)$  is arithmetic with  $d = 5$ .

**Investigate further:** This generalises: the sum of two arithmetic sequences is always arithmetic, with common difference equal to the sum of the two original common differences ( $2 + 3 = 5$  here). The same is not true for products, which is exactly what C3 tests next.

3. An AP  $(a_n)$  has common difference  $d$ . What can be said about  $b_n = a_n^2$ ?

- A)  $(b_n)$  is always arithmetic.
- B)  $(b_n)$  is always geometric.
- C)  $(b_n)$  is arithmetic only if  $d = 0$ .
- D)  $(b_n)$  is arithmetic only if  $a_1 = 0$ .

**Answer:** C) arithmetic only if  $d = 0$

**Method:**  $a_n = a + (n-1)d$ , so  $b_n = (a + (n-1)d)^2$  is a quadratic in  $n$  with leading coefficient  $d^2$ . A quadratic sequence is arithmetic (i.e. linear in  $n$ ) only if its leading coefficient vanishes:  $d^2 = 0 \Rightarrow d = 0$ .

**Investigate further:** This is the flip side of C2: sums of AP's stay linear in  $n$ , but squaring introduces an  $n^2$  term unless the original sequence was already constant.

4. An AP has all terms positive. The sum of the first 6 terms equals the sum of terms 7 through 10. Which must be true?

- A) The common difference  $d$  must be positive.
- B)  $d$  must be negative.
- C)  $d$  must be zero.
- D)  $d$  can have either sign.

**Answer:** A)  $d$  must be positive

**Method:**  $S_6 = 6a + 15d$ . Sum of terms 7–10 is  $4a + (6 + 7 + 8 + 9)d = 4a + 30d$ . Setting equal:  $6a + 15d = 4a + 30d \Rightarrow 2a = 15d \Rightarrow a = 7.5d$ . For  $a > 0$  (all terms positive requires at least  $a_1 > 0$ ),  $d$  must be positive too, since  $a = 7.5d$  forces  $d$  and  $a$  to share a sign.

**Investigate further:** Positivity of *all* terms is a stronger condition than just  $a > 0$  — with  $d > 0$  every later term is even bigger, so once  $a_1 > 0$  the rest follow automatically.

5. An AP has positive integer first term  $a$  and common difference  $d$  with  $a < d$ . Which pair  $(a, d)$  gives a sequence that contains the term 100 exactly?

- A) (5, 19)
- B) (6, 17)
- C) (7, 23)
- D) (8, 13)

**Answer:** A) (5, 19)

**Method:** Need  $d \mid (100 - a)$ . Check:  $100 - 5 = 95 = 19 \times 5 \checkmark$ ;  $100 - 6 = 94$ ,  $94/17$  is not an integer;  $100 - 7 = 93$ ,  $93/23$  is not an integer;  $100 - 8 = 92$ ,  $92/13$  is not an integer. Only A works.

**Investigate further:** This is the same divisibility idea as “does an AP hit a target value” in general:  $100 = a + (n - 1)d$  for some positive integer  $n$  exactly when  $d \mid (100 - a)$ .

6. An AP with first term  $a$  and common difference  $d$  satisfies  $S_{2n} = 2S_n$  for some particular  $n > 0$ . What does this force?

- A)  $d$  must equal 0.
- B)  $a$  must equal 0.
- C)  $n$  must be even.
- D) No constraint — this holds automatically for every AP.

**Answer:** A)  $d$  must equal 0

**Method:**  $S_{2n} = n(2a + (2n - 1)d)$  and  $2S_n = n(2a + (n - 1)d)$ . Setting them equal and dividing by  $n > 0$ :  $2a + (2n - 1)d = 2a + (n - 1)d \Rightarrow nd = 0$ . Since  $n > 0$ , this forces  $d = 0$ .

**Investigate further:** This is a genuinely surprising result: doubling the term count can only double the sum if the sequence isn’t actually changing at all. It’s the reason  $S_n$  growing quadratically (C7) matters — a truly arithmetic (non-constant) sequence’s sum grows faster than linearly, so doubling  $n$  more than doubles  $S_n$ .

7. For an AP with common difference  $d \neq 0$ , the sum  $S_n$  of the first  $n$  terms, viewed as a function of  $n$ , is:

- A) linear in  $n$ .
- B) quadratic in  $n$  with no linear term.
- C) quadratic in  $n$ .
- D) exponential in  $n$ .

**Answer:** C) quadratic in  $n$

**Method:**  $S_n = \frac{n}{2}(2a + (n-1)d) = \frac{d}{2}n^2 + (a - \frac{d}{2})n$ . Since  $d \neq 0$ , the  $n^2$  coefficient is nonzero, so  $S_n$  is genuinely quadratic (not linear, and it does have a linear term unless  $a = \frac{d}{2}$ ).

**Investigate further:** This is the single most useful fact in the whole sheet: any  $S_n$  you're handed that's a quadratic with no constant term secretly comes from an AP, and you can read off  $d$  and  $a$  directly from its coefficients — exactly what C8 and D4 ask you to do.

8.  $S_n = 3n^2 - n$  is the sum of the first  $n$  terms of an arithmetic sequence. Find its common difference.

- A) 3
- B) 6
- C) -1
- D) 2

**Answer:** B) 6

**Method:** Matching  $S_n = 3n^2 - n$  to  $S_n = \frac{d}{2}n^2 + (a - \frac{d}{2})n$ :  $\frac{d}{2} = 3 \Rightarrow d = 6$ , and  $a - \frac{d}{2} = -1 \Rightarrow a = 2$ .

**Investigate further:** The most common slip is answering 3 (forgetting the  $\frac{1}{2}$  in front of  $d$ ) or -1 (reporting the coefficient of  $n$  instead of solving for  $d$ ). Always solve  $\frac{d}{2} = (\text{coefficient of } n^2)$ , not  $d = (\text{coefficient of } n^2)$ .

## Section D Challenge

1. An AP has first term  $a$  and common difference  $d$ . The sum of the first 6 terms equals the sum of the first 11 terms. Which relationship between  $a$  and  $d$  does this force? (*after TMUA 2018 Paper 1 Q2*)

- A)  $a = -8d$
- B)  $a = 8d$
- C)  $a = -\frac{17}{2}d$
- D)  $a = \frac{17}{6}d$
- E)  $a = -4d$

**Answer:** A)  $a = -8d$

**Method:**  $S_6 = S_{11} \Rightarrow \frac{6}{2}(2a + 5d) = \frac{11}{2}(2a + 10d) \Rightarrow 3(2a + 5d) = 5.5(2a + 10d) \Rightarrow 6a + 15d = 11a + 55d \Rightarrow -5a = 40d \Rightarrow a = -8d$ .



**Investigate further:** The real TMUA question (2018 Paper 1 Q2) uses  $S_5 = S_8$  and lands on  $a = -\frac{38}{3}d$  — same technique, uglier numbers. Try that version by hand to see how the method scales.

2. A selection  $S$  of  $n$  terms is taken from the arithmetic sequence  $2, 5, 8, \dots, 80$ . What is the smallest value of  $n$  for which there must be two distinct terms in  $S$  summing to 82? (after TMUA 2022 Paper 2 Q8)
- A) 13  
B) 14  
C) 15  
D) 22  
E) 23  
F) 24

**Answer:** C)  $n = 15$

**Method:** The sequence  $2, 5, \dots, 80$  has 27 terms,  $a_i = 3i - 1$ . Two terms with indices  $i, j$  sum to 82 exactly when  $i + j = 28$ . Pairing indices 1–27 by this rule gives 13 genuine pairs  $(1, 27), (2, 26), \dots, (13, 15)$ , plus one unpaired index (14, whose partner would be itself). A selection can safely take one element from each of the 13 pairs plus the unpaired element — 14 terms with no pair summing to 82. Any 15th term forces a complete pair. So the smallest guaranteeing  $n$  is 15.

**Investigate further:** This is the Pigeonhole Principle in disguise: the “pairs” are the pigeonholes, and once you must place more elements than there are safe one-per-pair slots (plus the singleton), a full pair is forced. The real TMUA question (2022 Paper 2 Q8) uses the AP  $1, 4, \dots, 70$  and target sum 74, landing on  $n = 14$  — work through why the singleton index shifts between the two versions.

3. The first four terms of an arithmetic sequence  $p, p + d, p + 2d, p + 3d$  (with  $d > 0$ ) are all prime, and  $p > 3$ . Prove that  $d$  must be a multiple of 6.

**Answer:**  $d$  must be a multiple of 6.

**Method:** *Mod 2:* since  $p > 3$  is prime,  $p$  is odd. If  $d$  were odd,  $p + d$  would be even and greater than 2, hence composite — contradiction. So  $d$  is even.

*Mod 3:* since  $p > 3$ ,  $p \not\equiv 0 \pmod{3}$ . If  $d \not\equiv 0 \pmod{3}$ , then  $d$  is coprime to 3, so  $p, p + d, p + 2d$  take all three residues mod 3 (adding a unit repeatedly cycles through every residue) — meaning one of them is  $\equiv 0 \pmod{3}$ . That term is greater than 3 and divisible by 3, hence composite — contradiction. So  $3 \mid d$ . Combining:  $2 \mid d$  and  $3 \mid d \Rightarrow 6 \mid d$ .  $\square$

**Investigate further:** The smallest example is  $p = 5, d = 6$ : 5, 11, 17, 23 — all prime. Try to find the next one by hand, and notice how quickly  $d = 6$  stops being sufficient on its own (having  $6 \mid d$  is necessary, not sufficient, for four primes in AP).

4. An AP has  $S_n = An^2 + Bn$ . Given  $A = 5$  and that the 10th term of the sequence is 91, find  $B$ .
- A)  $-4$   
B)  $4$   
C)  $-9$   
D)  $9$

**Answer:** A)  $-4$

**Method:**  $a_n = S_n - S_{n-1} = A(2n-1) + B$  (expand and simplify  $An^2 + Bn - A(n-1)^2 - B(n-1)$ ).  
With  $A = 5$ :  $a_{10} = 5(19) + B = 95 + B = 91 \Rightarrow B = -4$ .

**Investigate further:** This formula  $a_n = A(2n-1) + B$  is worth memorising directly: it's the general rule for reading off any term of an AP straight from its  $S_n = An^2 + Bn$  form, without ever finding  $S_{n-1}$  by hand each time.

5. A sequence  $(a_n)$  (not assumed arithmetic) is such that its first differences  $a_{n+1} - a_n$  themselves form an arithmetic sequence in  $n$ . What can be said about  $a_n$  as a function of  $n$ ?

- A)  $(a_n)$  must itself be arithmetic.
- B)  $a_n$  is a quadratic function of  $n$ .
- C)  $(a_n)$  must be geometric.
- D) No general conclusion can be drawn.

**Answer:** B)  $a_n$  is a quadratic function of  $n$ .

**Method:** If the first differences  $a_{n+1} - a_n$  form an AP, then the *second* differences of  $(a_n)$  are constant — and a sequence with constant second differences is exactly a quadratic function of  $n$  (the discrete analogue of a constant second derivative meaning a parabola).

**Investigate further:** This is C7 in reverse: there, an AP's partial sum  $S_n$  turned out to be quadratic in  $n$  because  $S_n$ 's first differences ( $= a_n$ ) form an AP. This question generalises that observation to any sequence, not just partial sums — the same idea will resurface when this pillar looks at general recurrence relations.

## Top 5 Patterns — Worksheet #1

- nth-term formula:**  $a_n = a + (n-1)d$ . The foundation everything else builds on — always count *steps* (index differences), never raw indices. *(see A1, A2, A5, A9, A10)*
- Sum formula:**  $S_n = \frac{n}{2}(2a + (n-1)d) = \frac{n}{2}(a + l)$ . Use the  $(a + l)$  form whenever you're given the first and last term directly — it skips finding  $d$  entirely. *(see A3, A6, B1)*
- $S_n$  is quadratic in  $n$ , with no constant term.**  $S_n = \frac{d}{2}n^2 + (a - \frac{d}{2})n$  — read  $d$  and  $a$  straight off the coefficients, and extract any term via  $a_n = A(2n-1) + B$ . *(see C7, C8, D4, D5)*
- Middle-term trick for odd-length AP's.** When  $n$  is odd,  $S_n = n \times (\text{middle term})$  — no need to find  $a$  or  $d$  at all. *(see B4)*
- Pairing/Pigeonhole within an AP.** To force two terms to sum to a target, pair up indices by that target and count the leftover singleton — the answer is always (safe pairs) + (singletons) + 1. *(see D2)*

## Common Traps — Worksheet #1

- Forgetting the  $\frac{1}{2}$  in the sum formula.** Matching  $S_n = 3n^2 - n$  to  $\frac{d}{2}n^2 + \dots$  gives  $d = 6$ , not  $d = 3$  — always solve for  $d$  from the *halved* coefficient. *(see C8)*
- Trusting one example for a “must be true” claim.** A single valid  $(a, d, n)$  never proves a statement true — you need a counterexample to rule one out, or a full argument to confirm one

always holds. (see C1)

3. **Assuming every quadratic  $S_n$ -like formula is automatically a valid AP sum.** The  $n$ th-term extraction  $a_n = S_n - S_{n-1}$  only makes sense when  $S_0 = 0$  is consistent with the formula — check there's no stray constant term. (see D4)
4. **Sign errors when solving simultaneous AP equations.** A negative common difference is completely valid — don't discard a solution just because  $d < 0$  looks unusual. (see D1, B6)
5. **Declaring a sequence arithmetic from a few terms without proving it for general  $n$ .** Checking three consecutive differences match is suggestive, not a proof — squaring an AP (C3) is exactly the case where a plausible-looking pattern breaks down algebraically. (see C3)

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*“Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised.”*

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Found a mistake or want to contribute a sheet?  
Visit the open-source project at [github.com/The-CerealDev/speed-maths](https://github.com/The-CerealDev/speed-maths)